

AS • Edexcel • Physics

 2 mins  2 questions

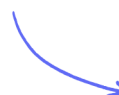
Multiple Choice Questions

Waves, Electrons & Photons

Diffraction / The Diffraction Grating Equation / Core Practical 8: Investigating Diffraction Gratings / The Wave Nature of Electrons / The de Broglie Equation / Transmission & Reflection of Waves / Pulse-Echo Technique / Wave-Particle Duality / Energy of a Photon

Easy (1 question)	/1
Medium (1 question)	/1
Total Marks	/2

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Easy Questions

- 1 Ultrasound can be used to investigate the structure of organs of the human body using the pulse-echo technique.

The level of detail obtained depends on the wavelength and the length of the pulses.

Which line of the table shows a change to wavelength and a change to pulse length that would each improve the level of detail?

	Wavelength	Pulse length
A	decrease	decrease
B	increase	decrease
C	decrease	increase
D	increase	increase

(1 mark)

Medium Questions

- 1 A monochromatic beam of light of wavelength λ from a laser is directed at a diffraction grating of line spacing d .

A student calculates the value of d/λ in order to determine the expected number of visible maxima.

The calculated value of d/λ is 4.7

How many maxima are visible?

- A. 4
- B. 5
- C. 9
- D. 11

(1 mark)